

8 (short) The mechanism shown in Fig. 7 is driven at joint A such that the member AB turns with angular velocity ω .

(a) Using the method of instantaneous centres, or otherwise, derive an expression for the velocity of the slider C. [5]

(b) If a friction force F acts at slider C and friction torques Q act at all joints A, B and C, find an expression for the torque needed to drive the mechanism. [5]

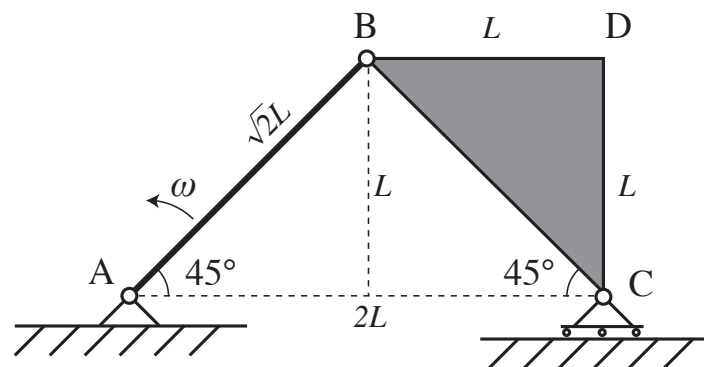


Fig. 7